

Towards Operational Conversational Intelligence: A Speech Intelligence Framework

Chaitanya Vishnoi*, Shudhant Khurana, Somya Rai, Saket Mohanty, Abhilash Timmapur

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Abstract

Body-worn camera (BWC) audio presents unique challenges—high ambient noise, variable recording conditions, and multiple overlapping speakers—that make automated transcription and speaker labeling very difficult. We propose a **dual-path conversational intelligence framework** that preprocesses raw BWC audio, splits the task into a diarization branch and an ASR branch, and fuses their outputs. The diarization branch uses a denoising front-end (DeepFilterNet) and voice activity detection (VAD) to identify speaker turns, followed by a multi-scale diarization model (MSDD) with TitaNet embeddings. The transcription branch uses loudness normalization and WhisperX (Large-v3) with forced alignment and probability-guided speech segmentation. Finally, we perform word-level speaker attribution by assigning each recognized word to the speaker segment with the maximum temporal overlap. We evaluate the proposed framework on a curated body-worn camera dataset constructed from publicly available U.S. and U.K. police body-worn camera recordings. Experimental results demonstrate that task-specific acoustic conditioning combined with probability-guided speech segmentation enables robust speaker diarization, reliable transcription, and accurate word-level speaker attribution under challenging body-worn camera recording conditions. The proposed modular architecture further provides an extensible foundation for future speaker-aware conversational intelligence systems.

Keywords: Body-worn camera, conversational intelligence, voice activity detection, speaker diarization, automatic speech recognition, WhisperX, Pyannote, DeepFilterNet, TitaNet, speaker attribution.

1 Introduction

Body-worn cameras (BWCs) are increasingly deployed by law enforcement agencies, generating large volumes of multimodal evidence from police–civilian interactions [1, 2]. Extracting meaningful information from these recordings through automatic transcription and speaker identification can significantly improve evidence review, officer training, incident analysis, and investigative workflows [2, 3].

However, BWC audio is considerably more challenging to process than conventional speech corpora. Recordings frequently contain environmental noise, overlapping speech, reverberation, microphone movement, sirens, traffic noise, radio communication, and rapidly varying recording conditions. These factors substantially degrade the performance of both automatic speech recognition (ASR)

*Primary author

and speaker diarization systems by reducing transcription accuracy and increasing speaker confusion [4, 5, 6].

Recent advances have produced highly capable models for the individual components of conversational speech processing. Whisper provides robust large-scale speech recognition, while WhisperX extends Whisper through voice activity detection and forced alignment to produce accurate word-level timestamps. Similarly, the Pyannote framework provides state-of-the-art neural voice activity detection and speaker diarization building blocks, and NVIDIA’s Multi-Scale Speaker Diarization Decoder (MSDD) achieves competitive speaker clustering through multi-scale speaker representations [7, 8, 9, 10]. Nevertheless, these models are typically optimized independently and are not designed as a unified processing pipeline specifically for noisy body-worn camera recordings.

To address this limitation, we propose an end-to-end **Conversational Intelligence** framework for BWC audio. Our approach follows a dual-path architecture in which the diarization and ASR branches employ different preprocessing strategies that are individually optimized for their respective tasks. The diarization branch performs DeepFilterNet-based speech enhancement followed by the proposed probability-guided Pyannote voice activity detection pipeline and multi-scale speaker diarization, whereas the ASR branch performs loudness normalization prior to WhisperX transcription and forced alignment. Finally, a deterministic maximum-overlap fusion algorithm combines the outputs of both branches by assigning each recognized word to the speaker segment exhibiting the greatest temporal overlap.

The proposed framework adopts a modular and reproducible architecture in which every processing stage produces auditable intermediate outputs. This enables independent evaluation of voice activity detection, speaker diarization, automatic speech recognition, and speaker attribution while facilitating future replacement or improvement of individual components without modifying the remainder of the pipeline.

The main contributions of this work are summarized as follows:

- A **modular end-to-end framework for speaker-aware conversational intelligence** from body-worn camera recordings, integrating voice activity detection, speaker diarization, automatic speech recognition, and speaker attribution.
- A **dual-path acoustic conditioning strategy** that independently optimizes the preprocessing requirements of speaker diarization and automatic speech recognition while maintaining a shared temporal grounding.
- A **probability-guided voice activity detection post-processing algorithm** that performs recursive speech segmentation based on speech posterior probabilities to generate robust WhisperX-compatible segments.
- A **deterministic word-level speaker-attribution** algorithm that integrates WhisperX forced alignment with speaker diarization using maximum temporal overlap.
- A comprehensive experimental evaluation on a manually annotated body-worn camera dataset, including quantitative analysis of voice activity detection, speaker diarization, automatic speech recognition, and speaker-attribution performance together with component-wise architectural ablation studies (Appendix B).

2 Research Gap and Motivation

Unlike controlled speech corpora, body-worn camera (BWC) recordings contain noisy far-field speech, overlapping conversations, highly variable acoustic environments, reverberation, microphone movement, and diverse speaking styles, making both automatic speech recognition and speaker diarization considerably more challenging [4, 5, 6, 3].

Furthermore, there is currently no large publicly available benchmark specifically designed for end-to-end conversational intelligence on police body-worn camera audio. Although several widely used speech corpora, including *LibriSpeech*, *AMI*, *CALLHOME*, *VoxConverse*, and *AISHELL*, have significantly advanced research in speech recognition and speaker diarization, they do not adequately represent the acoustic conditions encountered in operational BWC deployments [11, 12, 13, 14, 15].

This lack of publicly available benchmark datasets motivated the creation of our own evaluation dataset, comprising publicly available BWC-style recordings together with manually generated transcripts and speaker annotations. The dataset was designed to capture conversational characteristics and acoustic conditions commonly encountered in body-worn camera recordings while enabling stage-wise evaluation of the proposed framework.

Another practical challenge arises from the conflicting acoustic requirements of automatic speech recognition and speaker diarization. Aggressive speech enhancement can improve downstream speaker diarization by suppressing background interference, yet the same enhancement may distort acoustic cues that are important for robust speech recognition, thereby increasing transcription errors. In contrast, preprocessing strategies that preserve lexical information for ASR do not necessarily produce speaker representations that are optimal for diarization [16, 7, 8, 17, 10].

These observations motivate the proposed dual-path framework, where each branch employs task-specific preprocessing while preserving a common temporal representation that enables accurate fusion during the orchestration stage.

To the best of our knowledge, no existing open-source framework provides an end-to-end pipeline that jointly combines the components investigated in this work while specifically targeting conversational intelligence for body-worn camera recordings. We are unaware of an open-source framework that integrates task-specific acoustic conditioning, neural voice activity detection, speaker diarization, WhisperX transcription and forced alignment, and temporal speaker-attribution fusion into a unified pipeline specifically designed for body-worn camera audio [8, 9, 10, 16].

Several conversational and emotion-recognition datasets, including *IEMOCAP*, *MSP-Podcast*, and *MELD*, provide valuable multi-speaker speech recordings with rich annotations for emotion and dialogue analysis. However, these datasets were not designed for the acoustic conditions encountered in body-worn camera recordings, lacking characteristics such as severe environmental noise, radio communication, rapid speaker movement, and highly variable recording conditions. Consequently, they are unsuitable as end-to-end benchmarks for evaluating conversational intelligence pipelines in operational BWC scenarios [18, 19, 20].

3 Data

3.1 Data Collection

To evaluate the proposed framework, we assembled a small body-worn camera (BWC)-like dataset from publicly available English-language recordings. Since no standardized public benchmark

currently exists for end-to-end conversational intelligence on police body-worn camera audio, videos were collected from publicly accessible YouTube channels that publish body-worn camera footage, including official law enforcement agencies, government organizations, and publicly released police recordings [2, 1, 21].

The collected recordings were used exclusively for research and evaluation purposes, and all annotations were generated manually by the authors. Each recording was manually annotated to generate manually verified speaker-aware transcripts containing timestamped speaker turns. These annotations served as the ground truth for both ASR and speaker diarization evaluation.

An example annotation is shown below.

Speaker_0 [00:00:00 - 00:00:02] I think they're looking for us.
Speaker_1 [00:00:03 - 00:00:03] We are, yes.

The annotations were subsequently converted into RTTM format for diarization evaluation and word-level timestamp files for ASR assessment. To facilitate reproducible evaluation, the annotations were converted into standard formats required by the downstream evaluation tools, including RTTM files for speaker diarization and timestamped word-level references for automatic speech recognition.

3.2 Pre-processing

Prior to model inference, several audio conditioning strategies were investigated to determine their suitability for speaker diarization and automatic speech recognition. Conventional preprocessing techniques, including RMS normalization and peak normalization, were found to provide only limited improvements under challenging body-worn camera recording conditions. Consequently, the proposed framework employs task-specific acoustic conditioning tailored to the requirements of each processing branch.

For the diarization branch, DeepFilterNet was selected as the primary speech enhancement method owing to its ability to suppress environmental noise while preserving speech intelligibility [16]. Figure 1 compares the mel-spectrogram of a raw body-worn camera recording with its DeepFilterNet-enhanced counterpart. The enhanced spectrogram exhibits a noticeable reduction in low-frequency background noise and non-speech energy while preserving the dominant speech harmonics, providing a cleaner input signal for the subsequent speaker diarization pipeline.

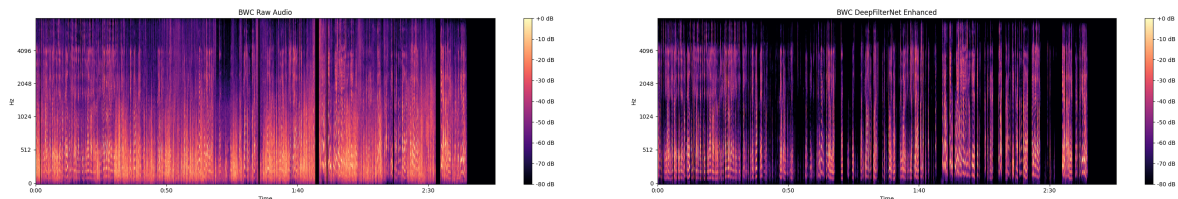


Figure 1: Mel-spectrogram comparison of a raw body-worn camera recording (left) and the corresponding DeepFilterNet-enhanced recording (right). DeepFilterNet suppresses low-frequency environmental noise while largely preserving the dominant speech components used for speaker diarization.

The conflicting behaviour observed during the preliminary experiments revealed what we refer to throughout this work as the **Enhancement Trap**. Although neural speech enhancement produces audio that is perceptually cleaner for human listeners, it simultaneously alters spectral and temporal

characteristics of the speech signal, including harmonic structure, transient information, and high-frequency phonetic cues. Modern ASR models, including Whisper, are trained on large-scale speech corpora containing substantial acoustic variability. Consequently, aggressive neural speech enhancement may alter acoustic characteristics that are informative for the ASR model, potentially reducing transcription accuracy despite improving perceptual speech quality. This behaviour was consistently observed during our experiments. In contrast, speaker diarization generally benefits from the suppression of background noise because cleaner speech provides more reliable inputs for speaker representation learning [17, 22] and speech activity detection. The observed trade-off between speaker diarization and automatic speech recognition forms the basis of the proposed dual-path architecture, in which speech enhancement is applied exclusively within the diarization branch, while the transcription branch preserves the original acoustic characteristics through loudness normalization alone [7, 8, 16]. Representative acoustic conditioning experiments supporting this design choice are presented in Appendix B.

Accordingly, the diarization branch operates on the DeepFilterNet-enhanced audio, whereas the transcription branch performs only loudness normalization following the ITU-R BS.1770 loudness normalisation recommendation [23] before WhisperX inference. Loudness normalization provides a consistent recording level without aggressively modifying the spectral characteristics of the speech signal, thereby preserving the information required for robust automatic speech recognition. Alternative speech enhancement approaches were also investigated during preliminary experimentation (see Appendix B); however, based on the observed trade-off between speaker diarization and transcription performance, DeepFilterNet and loudness normalization were selected as the final preprocessing strategies for the diarization and transcription branches, respectively.

Finally, all recordings were resampled to a single-channel 16 kHz waveform to ensure compatibility with the Pyannote voice activity detection and WhisperX transcription pipelines.

4 Architecture

Starting from a common preprocessed audio signal, the framework separates the processing pipeline into two independent branches, each optimized for a different downstream objective. Rather than forcing a single audio representation to satisfy both automatic speech recognition (ASR) and speaker diarization, the proposed architecture applies task-specific acoustic conditioning before recombining the outputs through a dedicated speaker-attribution module. This design is motivated by the **Enhancement Trap** discussed in Section 3.2. Representative acoustic conditioning experiments supporting this design choice are presented in Appendix B.

The first branch (Path A) is optimized for speaker diarization. The preprocessed waveform is enhanced using DeepFilterNet [16] to suppress environmental interference before passing through the proposed probability-guided Voice Activity Detection (VAD) pipeline. The resulting speech boundaries are subsequently supplied as externally generated VAD constraints to NVIDIA’s Multi-Scale Speaker Diarization Decoder (MSDD) [10], which employs pretrained TitaNet speaker embeddings [17] to generate timestamped speaker segments.

The second branch (Path B) is optimized for transcription. Instead of neural enhancement, the preprocessed waveform undergoes loudness normalization following the ITU-R BS.1770 recommendation before being processed by the same improved VAD pipeline. The detected speech segments are independently transcribed using WhisperX (Large-v3) [8], which additionally performs forced alignment to produce word-level timestamps and confidence scores.

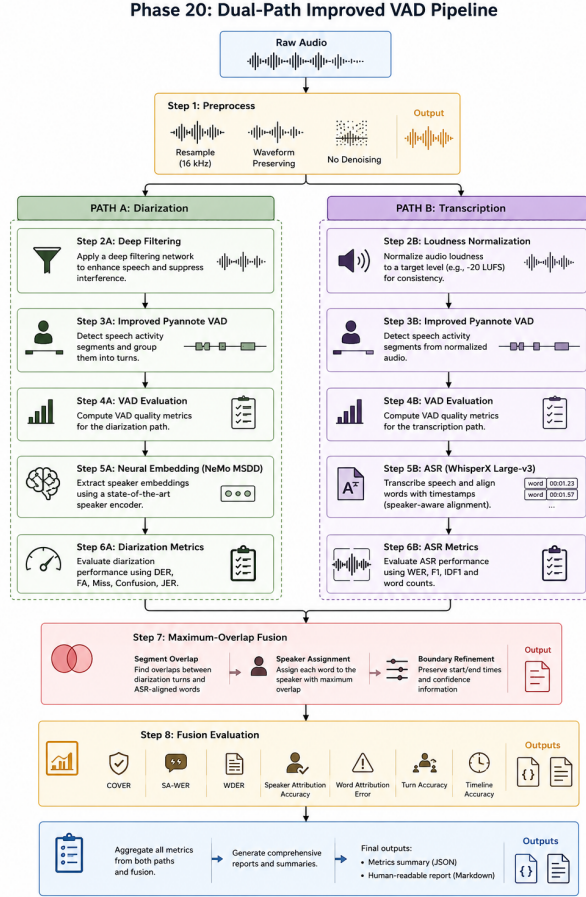


Figure 2: Overall architecture of the proposed dual-path framework for processing body-worn camera (BWC) recordings

Both branches operate on a common temporal grounding derived from the proposed VAD pipeline while preserving task-specific acoustic representations. This decoupling allows the diarization branch to benefit from aggressive noise suppression without introducing spectral artifacts into the transcription branch, thereby avoiding the conflicting optimization objectives commonly encountered when a single audio representation is shared between ASR and diarization. This modular design also allows improvements to either branch to be incorporated independently without modifying the remainder of the framework.

After both branches complete, the framework produces two complementary outputs: (i) a sequence of timestamped speaker diarization segments and (ii) a temporally aligned word-level transcription. These outputs are integrated through a deterministic speaker-attribution (orchestration) module, following the standard integration strategy adopted in modular ASR–speaker diarization pipelines [24]. For each transcribed word, the temporal overlap between the word interval and every diarization segment is computed, and the word is assigned to the speaker exhibiting the largest temporal overlap. If no temporal overlap exists, the nearest diarization segment within a predefined temporal tolerance is selected as a fallback. The resulting output is a speaker-attributed conversational transcript that preserves both the speaker identities estimated by the diarization branch and the word-level timestamps generated by the ASR branch.

The fusion stage produces both a machine-readable JSON representation containing word-level speaker attribution and a human-readable speaker-segmented conversational transcript. Finally, the framework performs comprehensive stage-wise evaluation, including VAD quality assessment, ASR evaluation, speaker diarization evaluation, fusion evaluation, and aggregate reporting, thereby enabling independent analysis of each component as well as the complete end-to-end system.

4.1 Acoustic Conditioning and Enhancement

The proposed dual-path architecture applies task-specific acoustic conditioning to the speaker diarization and automatic speech recognition (ASR) branches. Although both branches originate from the same repaired audio signal, they are optimized independently according to the requirements of their downstream tasks.

Although the acoustic conditioning differs between the two branches, both subsequently employ the proposed probability-guided Voice Activity Detection (VAD) pipeline to obtain a common temporal grounding for downstream speaker diarization and transcription.

The speaker diarization branch (Path A) processes the audio using DeepFilterNet prior to Voice Activity Detection (VAD) and speaker diarization. DeepFilterNet [16] performs neural speech enhancement by suppressing stationary and non-stationary background noise while preserving speech intelligibility. The enhanced waveform therefore exhibits improved speech-to-noise characteristics, providing cleaner inputs for the subsequent VAD and speaker diarization stages [17, 10] to operate on acoustically cleaner speech regions. This enhancement is particularly beneficial under the challenging acoustic conditions encountered in body-worn camera recordings, where environmental noise can substantially degrade speaker diarization performance.

In contrast, the transcription branch (Path B) preserves the original spectral characteristics of the speech signal. Rather than applying neural enhancement, the repaired audio is subjected only to loudness normalization using the ITU-R BS.1770 loudness recommendation [23] with a target integrated loudness of -20 LUFS. Formally, the normalized waveform is obtained as

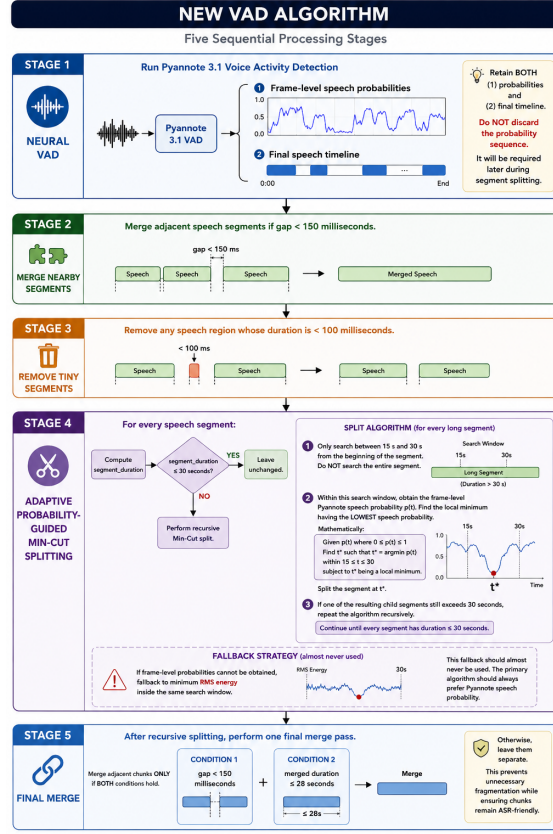
$$x_{\text{norm}}(t) = g_{\text{LUFS}} x(t)$$

where $x(t)$ denotes the preprocessed waveform and g_{LUFS} is the gain required to achieve the target integrated loudness. This normalization equalizes recording levels through a linear gain operation without modifying the spectral shape of the speech signal or introducing artificial enhancement artifacts.

This asymmetric conditioning strategy directly addresses the Enhancement Trap introduced in Section 3.2. Consequently, the proposed framework employs DeepFilterNet exclusively within the diarization branch while preserving the acoustically faithful normalized waveform for transcription. Representative experiments supporting this design choice are presented in Appendix B.

The resulting dual-path conditioning strategy allows each branch to operate on an acoustic representation that is specifically optimized for its downstream objective. Consequently, speaker diarization benefits from enhanced speech with more improved performance, whereas transcription preserves the original spectral characteristics required for accurate lexical recognition. This complementary design forms the acoustic foundation of the subsequent voice activity detection, speaker diarization, automatic speech recognition, and speaker-attribution stages of the proposed framework.

4.2 Voice Activity Detection



The proposed Voice Activity Detection (VAD) module forms the front-end for both the diarization (Path A) and transcription (Path B) branches. We employ the Pyannote neural Voice Activity Detector [9] to estimate frame-level speech probabilities over the input audio. Unlike standard Pyannote inference, where downstream pipelines typically operate on the binarized speech timeline for downstream processing, the proposed framework explicitly preserves both the continuous speech probability sequence and the corresponding binarized speech segments, enabling probability-guided post-processing during long-segment splitting.

Let

$$p(t) \in [0, 1]$$

denote the frame-level probability of speech predicted by Pyannote at time t . The neural predictions are internally binarized using Pyannote’s onset threshold, offset threshold, and hysteresis mechanism [9] to generate an initial speech timeline

$$G = \{G_i\}_{i=1}^N, \quad G_i = [t_i^{start}, t_i^{end}],$$

where each interval corresponds to a detected speech segment.

The proposed post-processing consists of five sequential stages, illustrated in Fig. 4.2. This probability-guided post-processing constitutes the principal methodological contribution of the proposed framework.

Stage 1 (Neural VAD). Pyannote performs frame-level speech activity estimation and produces both the speech probability sequence $p(t)$ and the initial binary speech timeline. Unlike the standard inference pipeline, our framework explicitly retains the frame-level probability sequence for subsequent probability-guided segmentation.

Stage 2 (Segment Merging). Adjacent speech segments are merged whenever the intervening silence duration is less than 150 ms. This prevents unnecessary fragmentation caused by brief pauses within continuous utterances while preserving genuine speaker pauses.

Stage 3 (Short Segment Removal). Speech regions shorter than 100 ms are discarded, as these segments are more likely to correspond to impulsive noise, transient artifacts, or isolated false-positive detections rather than meaningful speech.

Stage 4 (Adaptive Probability-Guided Min-Cut Splitting). Long speech segments are recursively divided to satisfy the maximum input duration supported by WhisperX. For every segment whose duration exceeds 30 s, a search window between 15 s and 30 s from the segment onset is examined. Within this interval, the algorithm searches for the local minimum of the retained speech probability sequence. Let

$$\mathcal{L} = \left\{ t \mid t_i^{\text{start}} + 15 \leq t \leq \min(t_i^{\text{start}} + 30, t_i^{\text{end}}), t \text{ is a local minimum of } p(t) \right\},$$

denote the set of all local minima within the search window. The optimal split point is then selected as

$$t^* = \arg \min_{t \in \mathcal{L}} p(t).$$

The speech segment is divided at t^* , thereby selecting the lowest-confidence speech region as the splitting point. If either resulting child segment still exceeds the maximum duration, the procedure is recursively applied until every segment satisfies the duration constraint.

Unlike the energy-based minimum-cut strategy adopted by [8], our approach directly utilizes the neural speech confidence estimated by Pyannote. Consequently, segmentation decisions are guided by the VAD model’s confidence rather than raw acoustic energy, which is expected to reduce sensitivity to environmental acoustic interference commonly encountered in body-worn camera recordings. Only when frame-level probability information is unavailable does the algorithm fall back to the conventional minimum-RMS-energy criterion.

Stage 5 (Final Merge). Following recursive splitting, a final merge pass is performed to eliminate unnecessary fragmentation. Two adjacent segments are merged only if (i) the silence gap between them is less than 150 ms and (ii) the merged segment remains shorter than 28 s. Otherwise, the segments remain separate. This final constraint preserves conversational continuity while ensuring compatibility with the subsequent WhisperX transcription stage.

The resulting speech segments serve as the common temporal grounding for both processing branches. For the diarization branch, the detected speech regions are converted into Oracle RTTM constraints [10] for NeMo MSDD. For the transcription branch, the corresponding temporal boundaries are applied to the loudness-normalized audio prior to WhisperX transcription and forced alignment. Consequently, both downstream branches operate on a consistent temporal segmentation while preserving task-specific acoustic representations, thereby simplifying the subsequent speaker-attribution process.

4.3 Speaker Diarization

The speaker diarization branch operates on the speech-enhanced audio together with the master grounding segments generated by the proposed Voice Activity Detection (VAD) pipeline. Rather than using NeMo’s internal MarbleNet-based Voice Activity Detector [10], the speech boundaries obtained from the proposed Pyannote-based VAD [9] are converted into an Oracle RTTM file and supplied directly to NVIDIA’s Multi-Scale Speaker Diarization Decoder (MSDD) [10]. Consequently, MSDD performs speaker embedding extraction, clustering, and speaker assignment only within the externally provided speech regions, ensuring that all downstream stages operate on a common temporal grounding.

Let the master grounding generated by the VAD be represented as

$$G = \{G_i\}_{i=1}^N, \quad G_i = [t_i^{start}, t_i^{end}],$$

where each interval corresponds to a detected speech segment. These intervals are converted into an Oracle RTTM file and used as the diarization constraints for MSDD, replacing the internal VAD stage while preserving the remaining diarization pipeline.

Prior to speaker embedding extraction, a binary speech mask derived from the master grounding is applied to the enhanced waveform. Unlike the default NeMo inference pipeline, speaker embedding extraction is therefore constrained exclusively to speech regions identified by the proposed VAD pipeline, preventing speaker embeddings from being extracted in regions identified as non-speech by the proposed VAD pipeline. Formally,

$$x_{\text{masked}}(t) = x(t) \cdot m(t),$$

where $x(t)$ denotes the enhanced audio signal and

$$m(t) = \begin{cases} 1, & t \in \bigcup_{i=1}^N G_i, \\ 0, & \text{otherwise.} \end{cases}$$

is the binary speech mask constructed from the VAD segments. This suppresses non-speech regions so that the speaker embedding extractor observes only active speech, thereby reducing the influence of background acoustic interference during downstream speaker diarization.

Speaker representation is obtained using NVIDIA’s TitaNet-Large speaker encoder [17], which generates fixed-dimensional speaker embeddings for each speech segment. These embeddings are subsequently processed by the NeMo Multi-Scale Speaker Diarization Decoder (MSDD) [10],

which jointly exploits multiple temporal resolutions to refine speaker segmentation and speaker assignments. The multi-scale formulation enables robust handling of rapid speaker transitions while maintaining consistent speaker identities throughout extended conversations. Representative experiments motivating the selection of the MSDD-based diarization pipeline are presented in Appendix B.4.

The diarization output is represented as a sequence of timestamped speaker segments,

$$D = \{(t_i^{start}, t_i^{end}, s_i)\}_{i=1}^M,$$

where s_i denotes the assigned speaker label.

Adjacent diarization segments assigned to the same speaker are merged whenever the temporal gap between consecutive segments does not exceed 2s, thereby reducing unnecessary fragmentation and producing coherent conversational turns for the subsequent word-level speaker attribution stage.

The final diarization output is exported in the standard RTTM format and subsequently combined with the word-level timestamps produced by WhisperX during the speaker-attribution fusion stage.

4.4 Speech-to-Text

The ASR branch transcribes the LUFS-normalized audio generated by Path B using the speech segments obtained from the improved VAD pipeline. We employ the WhisperX framework [8], together with OpenAI’s Whisper Large-v3 model [7], to perform automatic speech recognition. Each detected speech segment is transcribed independently with `condition_on_previous_text=False`, thereby preventing the transcription of one segment from influencing subsequent segments.

During preliminary experimentation, enabling contextual conditioning occasionally propagated transcription errors from acoustically challenging segments into subsequent speech segments. Since body-worn camera recordings frequently contain abrupt environmental disturbances and rapidly changing conversational contexts, disabling previous-text conditioning produced more stable and locally independent transcriptions for the proposed framework.

Let the improved VAD produce a sequence of speech segments

$$S = \{S_i\}_{i=1}^N, \quad S_i = [t_i^{start}, t_i^{end}],$$

where each segment satisfies the maximum duration constraint imposed by the VAD post-processing. For every speech segment, WhisperX produces

$$T_i = f_{ASR}(S_i) = \left\{ \left(w_j, t_j^{start}, t_j^{end}, c_j \right) \right\},$$

where w_j denotes the recognized word, t_j^{start} and t_j^{end} are the aligned word-level timestamps, and c_j represents the confidence score associated with the aligned word.

Following transcription, WhisperX performs forced alignment using language-specific phoneme alignment models [8] to refine the word-level timestamps. Forced alignment compensates for small segmentation offsets introduced during VAD and produces precise start and end times for every

recognized word. For example, the word “*vehicle*” may be aligned between 8.10 s and 8.55 s with an associated confidence score of 0.91.

The improved VAD post-processing ensures that every speech segment remains shorter than 30 s before transcription, thereby ensuring compatibility with the Whisper inference window for long-form recordings.

WhisperX combines VAD-guided segmentation with phoneme-level forced alignment and has demonstrated highly competitive performance for long-form automatic speech recognition and word-level timestamp estimation [8]. Within the proposed framework, WhisperX provides both the lexical transcription and the temporally precise word-level timestamps required by the subsequent speaker-attribution fusion stage. The confidence associated with every aligned word is preserved throughout the pipeline and included in the final speaker-attributed conversational transcript.

4.5 Speaker Attribution Fusion

The final stage of the proposed framework integrates the outputs of the speaker diarization and automatic speech recognition branches to generate a speaker-attributed conversational transcript. Following the deterministic orchestration strategy commonly adopted in modular ASR–speaker diarization pipelines [24], each transcribed word is assigned to the speaker whose diarization segment exhibits the greatest temporal overlap with the aligned word interval. If a word does not overlap any diarization segment, the temporally nearest speaker segment is selected as a fallback. This strategy preserves the precise word-level timestamps produced during forced alignment while combining the complementary strengths of the diarization and transcription branches.

Let the diarization output be represented as

$$D = \left\{ (t_i^{\text{start}}, t_i^{\text{end}}, s_i) \right\}_{i=1}^M,$$

where s_i denotes the speaker label associated with the diarization interval $[t_i^{\text{start}}, t_i^{\text{end}}]$. Similarly, let the aligned transcription be represented as

$$W = \left\{ (w_j, t_j^{\text{start}}, t_j^{\text{end}}, c_j) \right\}_{j=1}^N,$$

where w_j denotes the recognized word, t_j^{start} and t_j^{end} are the aligned word boundaries, and c_j is the corresponding confidence score.

For every transcribed word, the temporal overlap between the word interval and every diarization segment is computed. The assigned speaker label is obtained as

$$\hat{s}(w_j) = s_k, \quad k = \arg \max_i \left| [t_j^{\text{start}}, t_j^{\text{end}}] \cap [t_i^{\text{start}}, t_i^{\text{end}}] \right|,$$

where $|\cdot|$ denotes the duration of the overlapping interval. Consequently, every word is assigned to the speaker occupying the largest portion of its temporal duration. Since speaker attribution depends solely on temporal overlap, the proposed fusion algorithm is deterministic and does not require any additional learned parameters or post-processing models.

If a word does not overlap any diarization segment, the speaker label of the temporally nearest diarization segment is assigned as a fallback. If no suitable diarization segment exists, the word is assigned the label UNK. This strategy prevents isolated alignment inconsistencies from propagating through the conversational transcript while maintaining deterministic behaviour.

After speaker assignment, adjacent words are grouped into conversational turns, thereby reducing unnecessary speaker fragmentation. Consecutive words are merged whenever they belong to the same speaker and the temporal gap between adjacent words does not exceed 2 s. Formally, two consecutive words w_i and w_{i+1} are merged if

$$\hat{s}(w_i) = \hat{s}(w_{i+1})$$

and

$$t_{i+1}^{\text{start}} - t_i^{\text{end}} \leq 2 \text{ s.}$$

Otherwise, a new speaker turn is created. This procedure produces coherent conversational segments while preserving the original word-level timestamps.

The fusion stage generates two complementary outputs. The first is a machine-readable JSON representation containing every recognized word together with its aligned timestamps, confidence score, and assigned speaker label. The second is a human-readable conversational transcript in which adjacent words belonging to the same speaker are grouped into conversational turns. These complementary outputs provide both structured data for downstream conversational intelligence applications and an interpretable transcript suitable for manual review and subsequent language-model-based processing.

The quality of the generated speaker-attributed transcript is evaluated using the speaker attribution metrics described in Section 5. Speaker Accuracy measures the proportion of words correctly assigned to their corresponding speakers after optimal one-to-one speaker mapping. Word Attribution Error (WAE) is defined as the complement of Speaker Accuracy, while Timeline Accuracy measures the frame-level agreement between the predicted and reference speaker timelines. Together, these metrics quantify the quality of the final speaker-attributed transcript complementing the individual automatic speech recognition and speaker diarization evaluations.

5 Results

5.1 Experimental Evaluation

We evaluate the proposed dual-path conversational intelligence framework on an internally curated body-worn camera (BWC)-like dataset comprising publicly available recordings with manually annotated speaker-aware transcripts and diarization references. Each stage of the pipeline is evaluated independently using task-specific metrics before assessing the quality of the final speaker-attributed conversational transcript.

This stage-wise evaluation enables the contribution of each processing component to be analysed separately while also assessing the overall end-to-end performance. The reported metrics are obtained directly from the evaluation scripts described in Section 4 and are summarized in Tables 1, 2, 3, and 4. The numerical values reported below correspond to the proposed framework.

5.1.1 VAD

Table 1: Voice Activity Detection Performance

Metric	Path A	Path B
Precision	0.8448	0.8417
Recall	0.7721	0.7670
F1-score	0.8068	0.8026
False Alarm	0.2991	0.3042
Miss	0.2279	0.2330

Table 1 presents the VAD performance for both processing branches. The diarization branch (Path A) achieves an F1-score of 0.8068, while the transcription branch (Path B) achieves an F1-score of 0.8026. The nearly identical performance demonstrates that the proposed probability-guided post-processing remains stable under both acoustic conditioning strategies. Path A exhibits marginally higher precision and recall, suggesting that the enhanced audio provides slightly more robust speech detection under the evaluated recording conditions, without introducing excessive false alarms. Overall, the proposed VAD provides reliable speech segmentation for both downstream tasks while enforcing the maximum-segment-length constraint required by WhisperX.

The small performance difference between the two branches further indicates that the proposed probability-guided post-processing remains largely independent of the acoustic conditioning applied before VAD, allowing a common temporal grounding to be shared across both processing paths. Representative ablation experiments are presented in Appendix B.

5.1.2 Diarization

Table 2: Speaker Diarization Performance

Metric	Value
DER	0.4471
Miss	0.2300
False Alarm	0.1399
Confusion	0.0771
JER	0.8424

The proposed system achieves a Diarization Error Rate (DER) of 44.71%, comprising 23.00% Miss, 13.99% False Alarm, and 7.71% Speaker Confusion. The Miss and False Alarm components constitute the largest proportion of the overall DER, indicating that missed speech regions and false speech detections contribute more significantly to the total diarization error than incorrect speaker assignments. This suggests that, under the challenging acoustic conditions of body-worn camera recordings, accurately identifying speech activity remains a greater source of error than speaker discrimination once speech has been detected.

The Diarization Error Rate (DER) is computed as

$$\text{DER} = \text{Miss} + \text{FalseAlarm} + \text{Confusion},$$

where Miss denotes reference speech not detected, False Alarm corresponds to non-speech

incorrectly classified as speech, and Confusion measures speech assigned to the wrong speaker after optimal speaker mapping.

The corresponding Jaccard Error Rate (JER) is 0.8424. Unlike DER, which measures frame-level diarization errors, JER evaluates the temporal overlap between the predicted and reference speaker regions after optimal speaker matching. The relatively high JER indicates limited overlap between the predicted and reference speaker segments, reflecting the difficulty of achieving temporally accurate speaker segmentation in the presence of environmental noise, rapidly changing conversational turns, and overlapping speech. The selection of the final diarization pipeline is further justified in Appendix B.4.

5.1.3 ASR

Table 3: Automatic Speech Recognition Performance

Metric	Value
WER	0.3244
Substitutions	84
Insertions	27
Deletions	46
Reference Words	484
Hypothesis Words	465

Word Error Rate (WER) is computed as

$$\text{WER} = \frac{S + D + I}{N},$$

where S , D , and I denote the numbers of substitutions, deletions, and insertions, respectively, and N is the total number of reference words.

Table 3 reports the transcription performance of WhisperX using the loudness-normalized audio from Path B. The proposed system achieves a Word Error Rate (WER) of 32.44%, comprising 84 substitutions, 27 insertions, and 46 deletions over a reference transcript containing 484 words. The observed transcription errors primarily reflect the challenging acoustic characteristics of body-worn camera recordings, including environmental noise, overlapping speech, speaker movement, and conversational interruptions. Despite these adverse conditions, WhisperX successfully produces word-level transcriptions with temporally precise alignments, providing the timestamped lexical information required for the subsequent speaker-attribution fusion stage. These results further support the motivation behind the proposed dual-path architecture. During preliminary experimentation, applying neural speech enhancement to the transcription branch consistently increased transcription errors, whereas loudness normalization preserved the spectral characteristics required by WhisperX for robust lexical recognition. The final pipeline therefore performs transcription exclusively on the loudness-normalized audio while reserving speech enhancement for the diarization branch. The acoustic conditioning experiments motivating this design are discussed in Appendix B.2.

5.1.4 Fusion

Speaker Accuracy is computed as

Table 4: Fusion Performance

Metric	Value
Speaker Accuracy	0.9027
Word Attribution Error (WAE)	0.0973
Timeline Accuracy	0.7053

$$\text{SpeakerAccuracy} = \frac{1}{N} \sum_{i=1}^N \mathbf{1}(s_i^{\text{pred}} = s_i^{\text{ref}}),$$

where N denotes the total number of transcribed words after optimal one-to-one speaker mapping, s_i^{pred} and s_i^{ref} denote the predicted and reference speaker labels for the i -th word, respectively, and $\mathbf{1}(\cdot)$ is the indicator function. The proposed framework achieves a Speaker Accuracy of 90.27%, indicating that approximately nine out of every ten transcribed words are assigned to the correct speaker.

The Word Attribution Error (WAE) is defined as the complement of Speaker Accuracy,

$$\text{WAE} = 1 - \text{SpeakerAccuracy},$$

yielding a WAE of 9.73%, which indicates that relatively few words are incorrectly attributed following the proposed maximum-overlap fusion process.

Timeline Accuracy evaluates the temporal consistency of the generated conversational transcript by comparing the predicted and reference active-speaker sets at every 10 ms evaluation frame over the complete recording,

$$\text{TimelineAccuracy} = \frac{1}{K} \sum_{k=1}^K \mathbf{1}(\mathcal{A}_k^{\text{pred}} = \mathcal{A}_k^{\text{ref}}),$$

where K denotes the total number of evaluation frames, and $\mathcal{A}_k^{\text{pred}}$ and $\mathcal{A}_k^{\text{ref}}$ represent the predicted and reference sets of active speakers at frame k , respectively. A frame is counted as correct only when the complete active-speaker set exactly matches the reference. The proposed framework achieves a Timeline Accuracy of 70.53%, indicating that approximately seven out of every ten evaluation frames exhibit an exact match between the predicted and reference speaker activity. Since this metric jointly depends on the quality of speaker diarization, word-level alignment, and speaker attribution, it provides an end-to-end assessment of the temporal consistency of the generated conversational transcript.

Overall, the results indicate that the proposed maximum-overlap fusion strategy effectively integrates the complementary outputs of the diarization and transcription branches, producing a coherent speaker-attributed conversational transcript with high word-level speaker attribution accuracy while maintaining consistent temporal speaker alignment across the conversation.

5.2 Discussion

The experimental results validate the principal design decisions underlying the proposed dual-path conversational intelligence framework. The probability-guided VAD pipeline consistently produced reliable speech segmentation across both processing branches while enforcing the maximum segment duration required for WhisperX transcription. Because the same VAD framework is applied after branch-specific acoustic conditioning, both the speaker diarization and transcription pipelines operate on a common temporal grounding, simplifying the subsequent speaker-attribution process and reducing inconsistencies between independently generated outputs.

One of the key findings of this work is the differing influence of acoustic conditioning on speaker diarization and automatic speech recognition. Our preliminary experiments (see Appendix B) demonstrated what we refer to as the **Enhancement Trap**: although neural speech enhancement improves perceptual speech quality and facilitates speaker embedding extraction, it simultaneously modifies spectral characteristics that modern ASR models rely upon for robust lexical recognition. Consequently, applying DeepFilterNet to the transcription branch increased transcription errors, whereas loudness normalization preserved the acoustic characteristics required by WhisperX while still providing consistent recording levels. These observations support the proposed dual-path architecture, in which each branch is optimized independently according to its downstream objective rather than forcing a single audio representation to satisfy both tasks.

The deterministic maximum-overlap fusion algorithm further demonstrates the effectiveness of combining complementary information from the diarization and transcription branches. The high Speaker Accuracy and low Word Attribution Error indicate that accurate word-level speaker attribution can be achieved without introducing additional trainable models beyond the diarization and ASR components. Nevertheless, the lower Timeline Accuracy illustrates that the quality of the final conversational transcript remains fundamentally dependent on the temporal precision of both speaker diarization and word alignment. Errors introduced during voice activity detection, diarization, or transcription inevitably propagate through the subsequent fusion stage, highlighting the importance of improving each upstream component.

Although the proposed framework demonstrates promising performance on challenging body-worn camera recordings, several limitations remain. The evaluation was conducted on an internally curated dataset assembled from publicly available recordings, and therefore the reported results may not directly generalize to other recording conditions or operational deployments. Furthermore, the relatively high Diarization Error Rate indicates that accurately localizing speaker boundaries under highly dynamic acoustic conditions remains a challenging problem. Environmental noise, overlapping speech, rapid speaker transitions, and very short conversational turns continue to limit overall performance and represent important directions for future research.

6 Future Directions

6.1 LLM-based Speaker Attribution and Transcript Refinement

The current framework performs speaker attribution using a deterministic maximum-overlap fusion strategy, in which each aligned word is assigned to the speaker segment exhibiting the greatest temporal overlap. While this approach is transparent, computationally efficient, and fully reproducible, it relies exclusively on temporal information and does not exploit the semantic relationships that naturally exist within multi-speaker conversations.

Recent advances have demonstrated that Large Language Models (LLMs) have emerged as effective post-processing mechanisms for speaker diarization and speaker-attributed conversational understanding. Rather than replacing conventional speech processing components, these approaches leverage the outputs of existing automatic speech recognition and speaker diarization systems to reason about speaker identities using linguistic context, conversational structure, discourse coherence, and world knowledge.

For example, Wang *et al.* proposed *DiarizationLM* [24], in which an LLM refines speaker assignments after diarization by jointly analysing the conversational transcript and the initial speaker labels. Their results demonstrate that semantic reasoning can resolve speaker assignment errors that are difficult to distinguish using acoustic information alone, particularly during ambiguous speaker transitions and overlapping conversational exchanges.

Similarly, Park *et al.* [25] introduced a contextual beam-search framework that incorporates LLM reasoning during speaker assignment. Instead of relying solely on local acoustic evidence, the language model evaluates multiple speaker assignment hypotheses using conversational context, thereby improving diarization consistency across long-form conversations.

Beyond speaker diarization, Adedeji *et al.* [26] demonstrated that LLMs can also improve automatic speech recognition by correcting transcription errors through contextual reasoning. Although their work focuses on medical conversations, the underlying principle may also be applicable to body-worn camera recordings, where domain-specific terminology, challenging acoustic conditions, and fragmented conversational speech frequently reduce transcription quality.

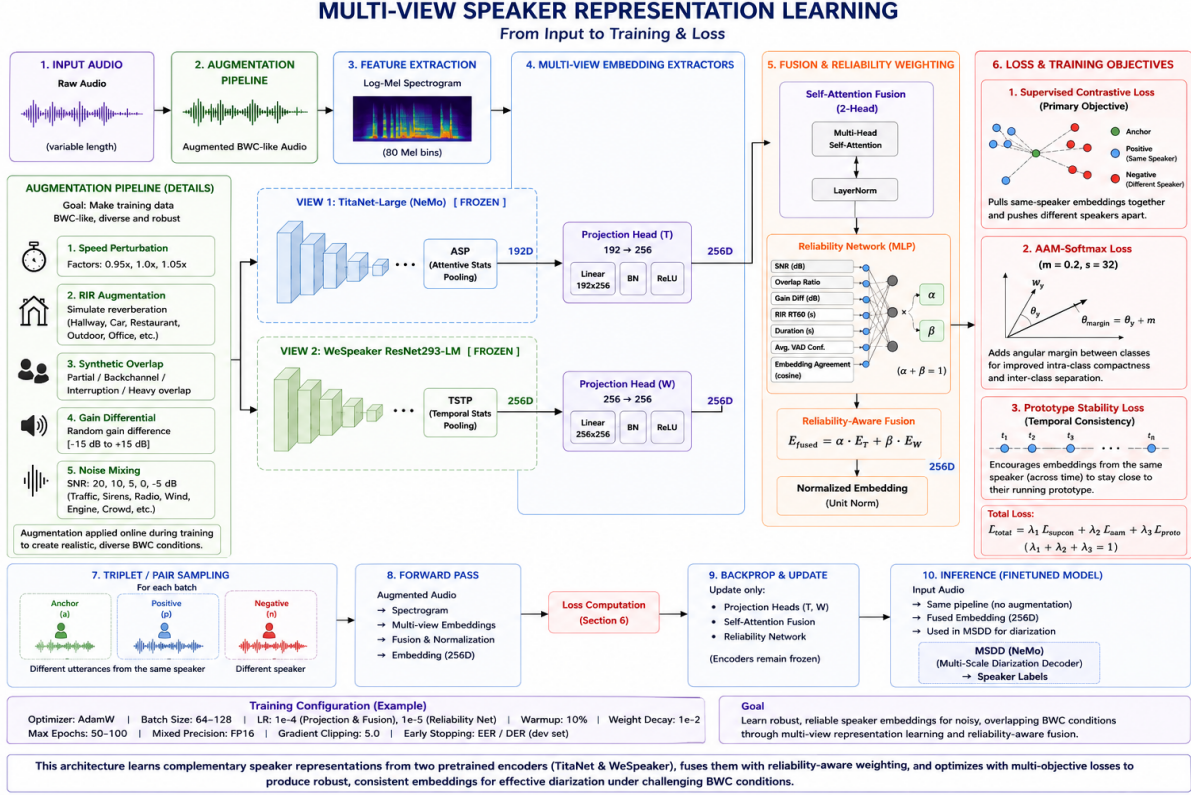
These recent developments suggest a natural extension of the proposed framework. Since the current pipeline already produces a temporally aligned speaker-attributed conversational transcript, an LLM could be incorporated as a modular post-processing stage after the speaker-attribution fusion module without modifying the underlying VAD, speaker diarization, or automatic speech recognition components. The language model would receive the attributed transcript together with the corresponding temporal information and refine speaker assignments using conversational semantics while preserving the temporal structure generated by the deterministic speaker-attribution module described in Section 4.5.

Beyond correcting residual speaker-attribution errors, such a semantic post-processing stage could enable higher-level conversational intelligence tasks, including dialogue summarization, speaker role identification (e.g., officer, civilian, witness), event extraction, question answering, automatic incident report generation, and conversational reasoning. Consequently, the proposed framework provides not only an end-to-end speech processing pipeline but also a structured conversational representation upon which future LLM-based conversational intelligence systems can be developed.

6.2 Multi-View Speaker Representation Learning

Although the proposed framework employs NVIDIA’s TitaNet speaker embeddings [17] within the MSDD diarization pipeline [10], the quality of speaker clustering remains fundamentally dependent on the robustness of the underlying speaker representation. In challenging body-worn camera recordings, factors such as environmental noise, reverberation, overlapping speech, speaker movement, and varying recording distances can significantly degrade the discriminative power of a single speaker embedding model.

Recent research has explored multi-view attention mechanisms for learning more robust speaker representations by capturing complementary local and global acoustic characteristics within Transformer-



based speaker embedding models [27]. Building upon this idea, a possible extension of the proposed framework is to employ multiple pretrained speaker encoders in their frozen state so that complementary acoustic representations can be extracted from the same speech segment. These embeddings could subsequently be projected into a shared representation space and fused using a lightweight self-attention mechanism to obtain a more robust speaker representation for downstream diarization.

Such multi-view learning strategies have also been shown to benefit from diverse acoustic augmentations during training. Multiple acoustic “views” of the same utterance can be generated using speed perturbation, room impulse response simulation, synthetic speaker overlap, gain variation, and realistic body-worn camera noise conditions, including traffic, wind, sirens, radio communication, engine noise, and crowd interference. Contrastive learning objectives are commonly used to encourage embeddings originating from the same speaker to remain close in the learned feature space while simultaneously maximizing the separation between different speakers.

A promising extension of the proposed framework is therefore to replace the current single-encoder speaker representation with a reliability-aware multi-view embedding module. Multiple pretrained speaker encoders, such as TitaNet [17], WeSpeaker [28], or similar architectures, could independently extract complementary speaker representations, which would then be projected into a common embedding space and fused through a lightweight self-attention mechanism. A reliability estimation network could further assign adaptive weights to each embedding based on acoustic characteristics such as signal-to-noise ratio, overlap ratio, reverberation, and voice activity confidence, allowing the fused representation to place greater emphasis on the most reliable speaker representation for each speech segment.

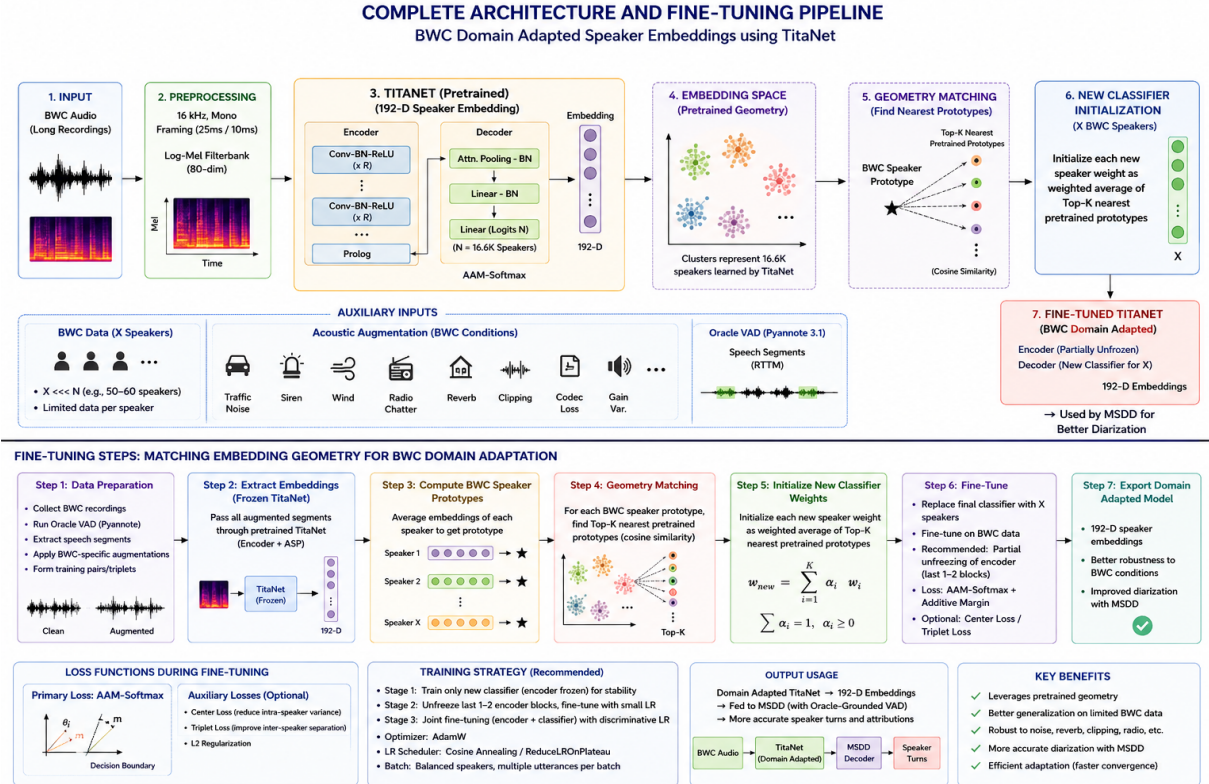
The resulting fused embedding could be integrated directly into the existing MSDD diarization

pipeline without modifying the remainder of the proposed architecture. Since the current framework already provides robust speech segmentation through the improved VAD pipeline, allowing the impact of improved speaker representations on downstream speaker diarization performance to be evaluated independently. Such a modular design preserves the advantages of the proposed dual-path framework while providing a principled direction for improving downstream speaker diarization under challenging body-worn camera recording conditions.

6.3 Task-Specific Speaker Embedding Adaptation

Unlike the previous subsection, which focuses on combining multiple pretrained speaker representations, the present direction investigates adapting a single pretrained speaker encoder to the body-worn camera domain.

The proposed framework currently employs the pretrained NVIDIA TitaNet speaker encoder [17] within the MSDD diarization pipeline [10] without additional domain-specific optimization. Although TitaNet demonstrates strong generalization across diverse speech datasets, it was originally trained on a large-scale speaker corpus containing approximately 16.6 thousand speakers under recording conditions that differ substantially from body-worn camera (BWC) environments. Consequently, the learned speaker embedding space may not fully capture the acoustic variability introduced by outdoor recordings, environmental noise, radio communication, reverberation, clipping, and rapidly changing speaker positions.



A promising direction for future work is to perform domain adaptation of the speaker encoder while preserving the discriminative geometry learned during large-scale pretraining. Rather than training a speaker encoder from scratch, the pretrained model can be fine-tuned using a comparatively small

corpus of labeled BWC recordings. For each speaker in the adaptation dataset, multiple utterances can first be used to estimate a stable prototype embedding, providing a reliable initialization for subsequent fine-tuning while maintaining the global structure of the pretrained embedding space.

During adaptation, the original classification layer can be replaced with a task-specific classifier corresponding to the speakers present in the adaptation dataset. Training can initially be restricted to the newly initialized classifier while the pretrained encoder remains frozen. Once the classifier converges, the final layers of the encoder can be progressively unfrozen and jointly optimized using a reduced learning rate. Such staged fine-tuning enables the encoder to gradually adapt to body-worn camera acoustics while reducing catastrophic forgetting of the speaker-discriminative representations learned during large-scale pretraining.

The adaptation process can be further strengthened through domain-specific data augmentation, including traffic noise, sirens, wind, radio communication, clipping, codec degradation, reverberation, gain variation, and synthetic speaker overlap. These augmentations can be combined with metric-learning objectives, such as Additive Angular Margin Softmax (AAM-Softmax) [29], Triplet Loss [30], or Center Loss [31], which are commonly used to encourage embeddings belonging to the same speaker to remain compact while simultaneously increasing the separation between different speakers under challenging acoustic conditions.

The resulting domain-adapted speaker encoder can be integrated directly into the proposed framework by replacing the pretrained TitaNet model within the MSDD diarization branch while leaving the remainder of the pipeline unchanged. Since the proposed architecture already provides consistent speech segmentation through the improved VAD pipeline and task-specific acoustic conditioning through the dual-path design, improving the underlying speaker representation offers a principled avenue for improving downstream speaker diarization performance for reducing speaker confusion and improving diarization performance while preserving the modular design of the overall system.

7 Conclusion

This paper presented a modular dual-path conversational intelligence framework for processing body-worn camera (BWC) audio under challenging real-world acoustic conditions. Recognizing the conflicting acoustic requirements of automatic speech recognition (ASR) and speaker diarization, the proposed framework independently is designed to optimise each task through dedicated processing branches while maintaining a common temporal representation for subsequent speaker attribution. The resulting architecture combines task-specific acoustic conditioning, an improved probability-guided Voice Activity Detection (VAD) pipeline, NeMo Multi-Scale Speaker Diarization Decoder (MSDD), WhisperX transcription with forced alignment, and a deterministic maximum-overlap fusion strategy to generate a temporally aligned, speaker-attributed conversational transcript.

A key contribution of this work is the introduction of a probability-guided VAD post-processing framework that preserves frame-level speech confidence throughout segmentation and uses this information to perform adaptive recursive min-cut splitting of long speech regions. Unlike conventional energy-based segmentation strategies, the proposed approach performs splitting directly on neural speech confidence estimates while simultaneously preserving a common temporal grounding for both the diarization and transcription branches. In addition, this work identifies what we refer to as the **Enhancement Trap**, demonstrating that speech enhancement and automatic speech recognition exhibit fundamentally different acoustic requirements. These observations motivate the proposed dual-path architecture, in which each branch receives an audio representation specifically optimized

for its downstream objective.

The modular design of the proposed framework enables every processing stage to generate auditable intermediate artifacts together with independent evaluation metrics. Voice activity detection, speaker diarization, automatic speech recognition, and speaker-attribution fusion are evaluated separately, providing detailed insight into the contribution and limitations of each component while facilitating reproducibility, debugging, and future experimentation. Furthermore, the modular architecture allows individual components to be replaced or improved without requiring changes to the remainder of the pipeline.

Experimental evaluation on an internally curated body-worn camera dataset indicates the effectiveness of the proposed framework. The improved VAD pipeline provides stable speech segmentation for both processing branches, while task-specific acoustic conditioning enables the diarization branch to benefit from enhanced speech and the transcription branch to preserve the acoustic characteristics required for robust lexical recognition. The proposed maximum-overlap fusion strategy further produces a speaker-attributed conversational transcript with approximately 90% Speaker Accuracy and a Word Attribution Error below 10%, demonstrating that complementary outputs from speaker diarization and automatic speech recognition can be effectively integrated into a coherent speaker-attributed conversational transcript.

Although the current evaluation is limited to an internally collected body-worn camera dataset, the proposed framework provides a flexible foundation for future research in conversational intelligence. The modular pipeline can naturally accommodate emerging advances in semantic reasoning using Large Language Models, multi-view speaker representation learning, and domain-adaptive speaker embedding optimization while preserving the overall system architecture. These extensions offer promising directions for improving robustness, speaker attribution, and higher-level conversational understanding in real-world body-worn camera recordings.

Overall, this work shows that combining task-specific acoustic conditioning, probability-guided speech segmentation, modular speaker diarization, robust transcription, and principled word-level speaker attribution provides an effective, interpretable, and reproducible framework for conversational analysis of body-worn camera audio. We believe the proposed framework establishes a strong baseline for future research in speaker-aware conversational intelligence and practical speech analytics for complex real-world acoustic environments.

By addressing the conflicting acoustic requirements of speaker diarization and automatic speech recognition through a unified yet modular architecture, the proposed framework contributes toward more reliable conversational intelligence for challenging body-worn camera recordings.

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A Baseline Pipeline Comparison

To place the proposed framework in context, we compare it with the the original single-path end-to-end pipeline used as the initial reference implementation. The baseline consisted of a conventional single-path processing pipeline comprising Silero Voice Activity Detection (VAD), WhisperX Large-v3 for transcription and forced alignment, Pyannote 3.1 speaker diarization, and the default WhisperX speaker-assignment procedure. In contrast, the proposed framework introduces task-specific acoustic conditioning through the dual-path architecture, probability-guided Pyannote VAD, Oracle-grounded NeMo Multi-Scale Speaker Diarization Decoder (MSDD) using TitaNet speaker embeddings, and a deterministic maximum-overlap speaker-attribution module.

Table 5: Comparison between the original baseline pipeline and the proposed framework.

Component	Baseline Pipeline	Proposed Framework
Audio Processing	Single-path	Dual-path (ASR / Diarization)
Speech Enhancement	None	DeepFilterNet (Path A only)
ASR Conditioning	Raw audio	−20 LUFS loudness normalization
Voice Activity Detection	Silero VAD	Pyannote VAD + probability-guided splitting
Speaker Diarization	Pyannote 3.1	Oracle-grounded NeMo MSDD + TitaNet
Speaker Attribution	WhisperX assignment	Maximum-overlap fusion
Evaluation	End-to-end only	Stage-wise evaluation (VAD, ASR, Diarization, Fusion)

Table 5 summarizes the principal architectural differences between the two systems. The proposed framework replaces the single shared audio representation with independent processing paths optimized for speaker diarization and automatic speech recognition. Furthermore, each stage of the pipeline produces intermediate artifacts together with dedicated evaluation metrics, enabling detailed analysis of the contribution of individual processing components.

Table 6: Quantitative comparison between the original baseline pipeline and the proposed framework.

Metric	Baseline	Proposed
VAD Precision	0.8250	0.8448
VAD Recall	0.8648	0.7721
VAD F1-score	0.8445	0.8068
DER	0.4599	0.4471
Speaker Confusion	0.1103	0.0771
WER	0.2907	0.3244
Speaker Accuracy	—	0.9027
Timeline Accuracy	—	0.7053

The comparison highlights an important design trade-off. While the proposed framework exhibits a modest increase in Word Error Rate relative to the original single-path baseline, it substantially reduces speaker confusion during diarization and introduces accurate word-level speaker attribution, timeline evaluation, and stage-wise analysis capabilities that were absent from the baseline implementation. These additional capabilities enable comprehensive conversational intelligence rather than transcription alone and therefore represent the primary objective of the proposed architecture.

Although the proposed framework does not uniformly improve every intermediate evaluation metric, it introduces several capabilities that are absent from the baseline pipeline, including

probability-guided speech segmentation, Oracle-grounded neural speaker diarization, explicit word-level speaker attribution, timeline-level conversational evaluation, and modular stage-wise analysis. These additions transform the baseline speech processing pipeline into a complete conversational intelligence framework while preserving reproducibility and facilitating future extensions.

B Component-wise Ablation and Design Justification

The final framework presented in this paper was derived through a series of controlled experiments investigating acoustic conditioning, voice activity detection, speaker diarization, and speaker-attribution strategies for body-worn camera (BWC) recordings. Rather than resulting from a single design choice, the proposed architecture emerged through successive evaluation of alternative processing strategies under realistic recording conditions.

This appendix summarizes only those experiments that directly influenced the final architectural decisions. Implementation details and exploratory experiments that did not directly influence the final architecture are intentionally omitted.

B.1 Evolution of Proposed Framework

Figure 3 summarizes the evolution of the proposed framework. Beginning from a conventional single-path speech processing pipeline, the framework progressively incorporated task-specific acoustic conditioning, oracle-grounded speaker diarization, probability-guided voice activity detection, and deterministic speaker-attribution fusion.

The principal experimental milestones are summarized below.

1. **Acoustic conditioning.** Initial experiments evaluated several preprocessing strategies, including conventional normalization techniques, DeepFilterNet-based enhancement, and SpeechBrain enhancement, in order to determine their influence on automatic speech recognition and speaker diarization.
2. **Speaker diarization.** The performance of conventional Pyannote-based diarization was subsequently compared with Oracle-grounded NeMo Multi-Scale Speaker Diarization Decoder (MSDD), allowing the influence of externally supplied speech boundaries to be isolated.
3. **Dual-path processing.** The conflicting behaviour observed between speech enhancement and transcription motivated the introduction of separate processing branches for automatic speech recognition and speaker diarization.
4. **Probability-guided speech segmentation.** The final architectural refinement introduced probability-guided recursive segment splitting, allowing the temporal segmentation produced by Pyannote to satisfy WhisperX duration constraints while preserving low-confidence transition regions.
5. **Speaker attribution.** The final conversational transcript is generated through deterministic word-level maximum-overlap fusion, replacing earlier segment-level assignment strategies explored during development.

Collectively, these studies indicate that the final architecture did not arise from a single architectural modification but rather from a sequence of component-wise investigations. Each major design decision was retained only after experimental evaluation indicated that it addressed a specific limitation identified in the preceding configuration. The resulting framework therefore reflects a

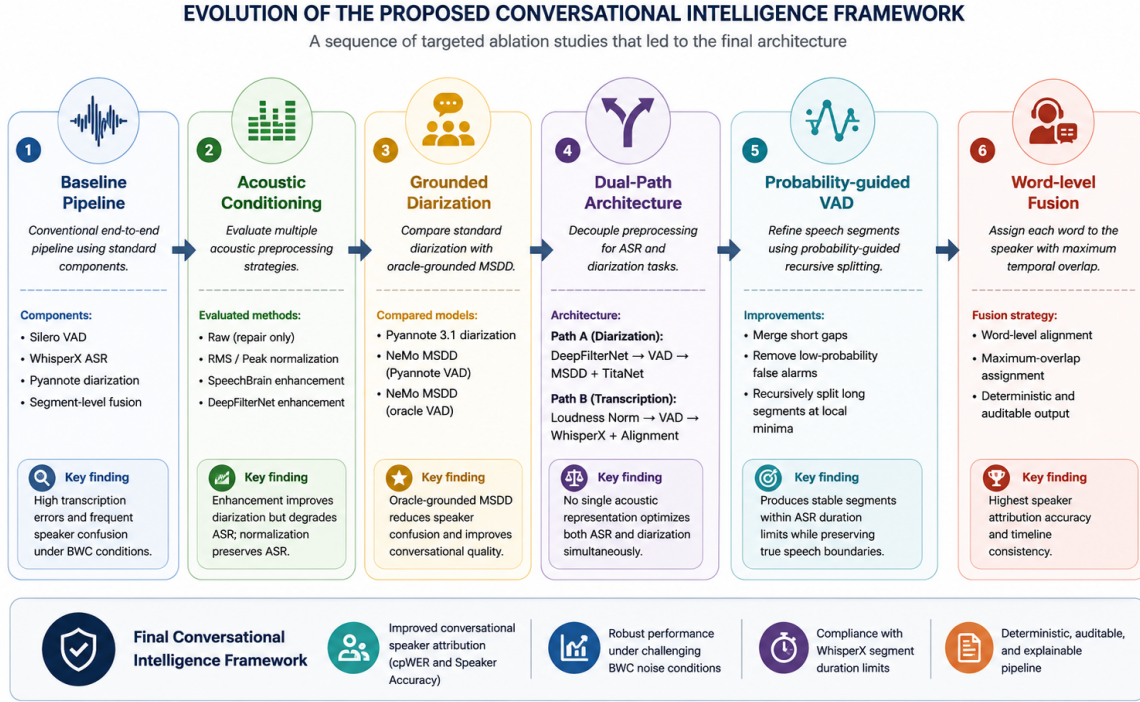


Figure B.1: Evolution of the proposed framework. Each stage represents the outcome of targeted ablation studies that addressed a specific limitation of the preceding pipeline and contributed to the final architecture.

Figure 3: Evolution of the proposed framework. Only the principal architectural milestones that directly contributed to the final system are shown. Each milestone represents the outcome of a controlled experimental evaluation rather than an implementation iteration.

combination of experimentally justified design choices rather than a single end-to-end architectural redesign.

B.2 Acoustic Conditioning Ablation

The first component investigated during the development of the proposed framework was acoustic conditioning. Since both speaker diarization and automatic speech recognition operate directly on the input waveform, the choice of preprocessing strategy has a significant influence on downstream performance. Conventional speech processing pipelines typically assume that improving perceptual speech quality benefits all subsequent tasks equally. However, our preliminary experiments demonstrated that this assumption does not hold for body-worn camera (BWC) recordings.

To investigate this behaviour, several preprocessing strategies were evaluated, including loudness normalization, DeepFilterNet neural speech enhancement, and SpeechBrain enhancement. The objective was to determine whether a single acoustic representation could simultaneously optimize automatic speech recognition and speaker diarization.

Table 7: Representative acoustic conditioning experiments performed during framework development.

Configuration	WER	DER	Observation
Normalized baseline	0.2975	0.4114	Provided the best balance between transcription and speaker diarization while preserving the original acoustic characteristics required for ASR.
DeepFilterNet (global)	0.3698	0.4408	Improved downstream speaker diarization but noticeably increased transcription errors when applied throughout the complete pipeline.
SpeechBrain enhancement	0.4504	0.5603	Produced the largest degradation in both transcription and diarization performance.
Full DeepFilterNet pipeline	0.4215	0.6267	Global enhancement proved unsuitable for end-to-end conversational intelligence despite producing perceptually cleaner speech.

The experiments consistently revealed a trade-off between transcription accuracy and speaker diarization performance, referred to throughout this work as the **Enhancement Trap**. Although neural speech enhancement produced speech that was perceptually cleaner for human listeners, applying the enhanced waveform throughout the complete processing pipeline consistently increased transcription errors while altering downstream speaker diarization behaviour. These observations indicate that improvements in perceptual speech quality do not necessarily translate into improvements in downstream machine perception.

The observed trade-off indicates that no single acoustic representation is equally suitable for both downstream tasks. While DeepFilterNet effectively suppresses environmental interference, the resulting acoustic modifications were empirically observed to alter the behaviour of downstream speaker diarization and automatic speech recognition. Conversely, loudness normalization preserves the original spectral structure of the speech signal while providing a consistent recording level for automatic speech recognition.

These observations directly motivated the task-specific acoustic conditioning strategy adopted in the final framework. DeepFilterNet is therefore applied exclusively within the speaker diarization branch, whereas the transcription branch operates on the loudness-normalized waveform. By allowing each branch to operate on an acoustic representation better suited to its respective objective, the proposed dual-path architecture resolves the observed conflict between speaker diarization and automatic speech recognition while preserving a common temporal grounding through the shared VAD pipeline.

B.3 Voice Activity Detection Design Study

Following the selection of task-specific acoustic conditioning, the next stage of development focused on improving the robustness of voice activity detection (VAD). Since both speaker diarization and automatic speech recognition depend on accurate speech segmentation, errors introduced during VAD inevitably propagate through the remainder of the conversational intelligence pipeline. Consequently, several alternative VAD strategies were evaluated to determine their suitability for long-form body-worn camera recordings.

The initial implementation employed Silero VAD owing to its lightweight design and real-time inference capability. Although Silero produced competitive frame-level detection accuracy, qualitative

inspection revealed that long speech segments were frequently fragmented, while abrupt environmental noise occasionally introduced false-positive speech regions. These observations motivated the adoption of the neural Pyannote VAD model, which provides frame-level speech posterior probabilities rather than binary speech decisions alone.

Unlike conventional inference pipelines, the proposed framework explicitly retains these probability estimates after speech detection. This enables the recursive probability-guided splitting strategy introduced in Section 4.3, allowing long speech segments to be divided at local minima of the speech posterior rather than according to fixed-duration windows or minimum-energy criteria.

Table 8: Representative comparison between the initial Silero-based speech grounding and the final probability-guided VAD pipeline.

Metric	Silero Baseline	Proposed VAD
Precision	0.8250	0.8448
Recall	0.8648	0.7721
F1-score	0.8445	0.8068
Miss Rate	0.1352	0.2279
False Alarm Rate	0.3971	0.2991

Table 8 highlights that the proposed VAD was designed to improve the quality of the temporal grounding provided to downstream modules rather than to maximize conventional frame-level VAD metrics alone.

Although the proposed probability-guided VAD does not improve every conventional frame-level evaluation metric, the experiments reveal an important trade-off. Compared with the initial Silero-based implementation, the final VAD pipeline reduces the false alarm rate (39.71% to 29.91%) while increasing precision (82.50% to 84.48%). These improvements are accompanied by lower recall, resulting in a more conservative speech detector. For conversational intelligence applications, reducing false-positive speech segments is particularly beneficial because false-positive speech regions propagate directly into both the speaker diarization and transcription branches, increasing the likelihood of speaker confusion and transcription errors.

More importantly, the objective of the proposed VAD is not solely to maximize frame-level detection accuracy. In contrast to conventional VAD pipelines, the proposed framework retains the frame-level speech posterior estimated by Pyannote and employs probability-guided recursive splitting to generate speech segments that simultaneously satisfy WhisperX duration constraints and preserve a common temporal grounding for both processing branches. Consequently, the VAD module serves not only as a speech detector but also as the source of the common temporal grounding shared by both processing branches.

It is also important to note that the probability-guided recursive splitting introduced in the final framework was motivated primarily by exploratory experiments rather than by the evaluated BWC recording itself. Earlier experiments identified voice activity detection as a major bottleneck under challenging recording conditions, particularly when long speech segments were present. Consequently, the proposed post-processing was designed to improve the robustness and reproducibility of speech segmentation even when no measurable difference is observed for a specific recording. The final framework therefore retains the probability-guided segmentation strategy as a deterministic safeguard rather than solely as a mechanism for improving conventional VAD metrics.

B.4 Speaker Diarization Ablation

The final framework employs NVIDIA’s Multi-Scale Speaker Diarization Decoder (MSDD) together with TitaNet speaker embeddings rather than the standard Pyannote speaker diarization pipeline. This design decision emerged from a series of controlled experiments investigating whether improvements in conversational speaker attribution necessarily correspond to reductions in frame-level diarization error.

Early experiments adopted the Pyannote speaker diarization pipeline owing to its strong performance on public benchmarks and ease of deployment. Although Pyannote consistently achieved competitive Diarization Error Rate (DER), error analysis revealed that conversational speaker attribution remained limited by speaker confusion, particularly during rapid conversational exchanges and overlapping speech. These observations motivated the investigation of NeMo’s oracle-grounded MSDD architecture, which combines multi-scale TitaNet speaker embeddings with a neural decoder capable of modelling speaker transitions across multiple temporal resolutions.

The objective of these experiments extended beyond minimizing frame-level diarization error. Since the ultimate goal of the proposed framework is to produce an accurate speaker-attributed conversational transcript, the experiments focused on identifying the architectural factors responsible for stable speaker segmentation under challenging body-worn camera recording conditions.

Table 9: The reported metrics summarize the principal design decisions that led to the final diarization architecture.

Configuration	DER	Speaker Confusion	Principal Observation
Pyannote baseline	0.411	0.110	Competitive frame-level diarization performance, but speaker confusion during rapid speaker transitions limited conversational consistency.
Oracle-grounded MSDD	0.449	0.078	Using externally generated speech boundaries substantially reduced speaker confusion despite a slight increase in overall DER.
Final framework	0.447	0.077	Selected as the final diarization architecture owing to reduced speaker confusion while maintaining competitive overall diarization performance.

An important observation emerging from these experiments is that minimizing Diarization Error Rate (DER) alone does not necessarily produce the most useful speaker segmentation for conversational intelligence. Error analysis revealed that speaker confusion during rapid speaker transitions and overlapping speech had a disproportionate impact on the quality of the final conversational transcript. Consequently, the design objective shifted from minimizing DER in isolation toward producing temporally stable speaker segments.

Additional exploratory experiments further indicated that the quality of the voice activity detection front-end strongly influences downstream speaker diarization behaviour. Supplying externally generated speech boundaries consistently reduced speaker confusion, indicating that the quality

of the speech segmentation has a substantial influence on the overall diarization pipeline under challenging body-worn camera recording conditions.

These observations directly motivated the final diarization architecture adopted in this work. Rather than replacing the diarization backend, the proposed framework focuses on improving the quality of the temporal speech grounding provided to MSDD through the probability-guided VAD pipeline while retaining TitaNet speaker embeddings and the multi-scale speaker decoder. This modular design enables future improvements in speech segmentation or speaker embedding learning to be incorporated independently without modifying the remainder of the framework.

Overall, the speaker diarization experiments demonstrate that the final framework was designed to improve robust speaker segmentation for speaker-attributed conversational intelligence rather than to minimize frame-level diarization metrics alone. The selection of oracle-grounded MSDD with TitaNet embeddings therefore reflects a design choice motivated by robustness, modularity, and compatibility with the proposed dual-path architecture.

B.5 Dual-Path Architecture Ablation

The preceding ablation studies demonstrated that the acoustic representation most suitable for speaker diarization differs from that required for automatic speech recognition. Neural speech enhancement and loudness normalization exhibited conflicting behaviour across the two downstream tasks, motivating the development of the proposed dual-path architecture.

Several pipeline configurations were subsequently evaluated to determine whether separating the preprocessing strategy for speaker diarization and automatic speech recognition could resolve the conflicting behaviour observed during the acoustic conditioning experiments.

Table 10: Architectural comparison of the preprocessing strategies investigated during framework development.

Configuration	Preprocessing Strategy	Principal Observation
Single-path	Identical preprocessing for ASR and diarization	Simplifies the pipeline but forces both tasks to operate on the same acoustic representation.
Global enhancement	DeepFilterNet applied before both ASR and diarization	Reduced background interference but consistently degraded transcription accuracy.
Global normalization	Loudness normalization before both ASR and diarization	Maintained reliable transcription but provided less favourable conditions for speaker diarization.
Dual-path architecture	DeepFilterNet for diarization and loudness normalization for ASR	Allows each downstream task to operate on an acoustic representation better suited for its respective objective while maintaining common temporal grounding.

These observations directly motivated the separation of the pipeline into two independent processing branches. The diarization branch operates exclusively on speech-enhanced audio in order to maximize speaker discriminability, whereas the transcription branch preserves the original

spectral characteristics through loudness normalization before WhisperX inference. Both branches subsequently share a common temporal grounding generated by the proposed probability-guided voice activity detection pipeline.

The proposed architecture also improves modularity. Since the diarization and transcription branches become independent immediately after the shared VAD stage, future improvements in speech enhancement, speaker diarization, or automatic speech recognition can be incorporated into either branch without requiring modifications to the remainder of the framework.

Overall, the dual-path architecture emerged as the natural consequence of the preceding ablation studies rather than as an independently motivated design choice. By allowing speaker diarization and automatic speech recognition to operate on task-specific acoustic representations while preserving a common temporal grounding, the proposed framework addresses the conflicting optimization requirements observed throughout the acoustic conditioning experiments and provides a modular foundation for future conversational intelligence systems.